Theorem Proof Delivery

Expert inference validation report

Focus question
Do the obligations and lemma chain establish the exact theorem?

Run Id
fixture.theorem_proof_delivery.base

Target
mathematical

Trace Id
trace.009c88ef320336eb

Workflow Id
theorem_proof_delivery

Workflow Name
Theorem Proof Delivery

Workflow Version
1.0.0

Canonical packet SHA-256
607ba13348ec6b13a20c3c1d9f8ae1bf1ff64fe91916e256e3d4ba67498eb201

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1 Expert reading contract

Public rationale boundary

Inspectable public rationale and evidence only. The inventory must not contain private chain-of-thought or derive scientific claims from layout, chronology, control authorization, or artifact integrity.

This document records inspectable premises, public warrants, evidence, conclusions, uncertainty, scope, authority, failures, and obligations. It does not expose private chain-of-thought. Execution authorization, artifact integrity, scientific epistemic status, and workflow disposition are independent. Chronological precedence does not mean logical or evidential support.

1.1 Status legend

Artifact integrity: whether referenced bytes and receipts validate **Scientific epistemic status:** the bounded support assigned to a claim **Workflow disposition:** what the workflow did with the item

2 Validation dashboard

Control coverage

1.0

Semantic coverage

1.0

Evidence resolution

1.0

Visible trace gaps

0

Required expert contrasts

["exact theorem versus proved statement", "lemma support versus circularity", "closed obligation versus gap or countermodel"]

3 Declared route and topology

Route spine

Statement to proof gate

Topology mode

dag

Step	Skill	Mode / observed state	Consumes	Produces
orchestrate_delivery	proof-delivery-workflow-orchestrator	required / observed	["ProofQueueItem"]	["ProofDeliveryPlan"]
formalize_obligations	proof-obligation-formalizer	required / observed	["ProofDeliveryPlan"]	["FormalProofObligations"]
build_lemma_chain	lemma-chain-proof-builder	required / observed	["FormalProofObligations"]	["LemmaChain"]
split_gap_or_countermodel	proof-gap-countermodel-splitter	conditional / observed	["LemmaChain"]	["ProofGapDecision"]

Step	Skill	Mode / observed state	Consumes	Produces
control_pvsnp_inference	pvsnp-inference-quality-controller	conditional / observed	["LemmaChain"]	["InferenceQualityDecision"]
gate_proof	proof-quality-gatekeeper	validator_only / observed	["LemmaChain"]	["ProofGateDecision"]

4 Profile topology and workflow-specific contrasts

Projection lens

theorem_proof_delivery

Topology mode

dag

Required expert contrasts

["exact theorem versus proved statement", "lemma support versus circularity", "closed obligation versus gap or countermodel"]

Declared profile topology

No specialized topology applies

Observed specialist roles

No specialist fan-out applies

A specialist fork or join is a workflow relation, not proof that specialist conclusions are mutually consistent. Finite, endpoint, asymptotic, and continuum boundaries remain separate unless an explicit validated inference connects them.

5 Expert inference dossiers

Each dossier is bound to its semantic event ID. Control-only events appear in the lifecycle appendix and are not scientific claims.

5.1 Inference 1: theorem_proof_delivery.base.orchestrate_delivery.inference

---

Trace disposition

Complete public dossier

Step / skill

orchestrate_delivery / proof-delivery-workflow-orchestrator

Question

At orchestrate_delivery, how does ProofQueueItem bear on ProofDeliveryPlan? Does the lemma DAG prove the exact quantified statement without circular support?

Typed premises

[{"statement": "ProofQueueItem: The exact theorem statement and formal obligations are source-bound.", "type": "artifact"}]

Transformation / operator

validate_exact_claim_boundary

Bounded public warrant

At orchestrate_delivery, validate_exact_claim_boundary checks ProofQueueItem before emitting ProofDeliveryPlan. Authority boundary: May validate only the exact bound statement after the proof gate; may not silently close gaps or widen quantifiers, model, or regime.

Conclusion

orchestrate_delivery produces ProofDeliveryPlan: Quarantine the proof pending the endpoint lemma.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}

Causal parents

["theorem_proof_delivery.base.orchestrate_delivery.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.5dba134bef7cd8

Alternative / falsifier / failure links

alternative.fc402c24a799fd, falsifier.4b5d8d42fecd8b

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["orchestrate_delivery-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.2 Inference 2: theorem_proof_delivery.base.formalize_obligations.inference

---

Trace disposition

Complete public dossier

Step / skill

formalize_obligations / proof-obligation-formalizer

Question

At formalize_obligations, how does ProofDeliveryPlan bear on FormalProofObligations? Does the lemma DAG prove the exact quantified statement without circular support?

Typed premises

[{"statement": "ProofDeliveryPlan: The exact theorem statement and formal obligations are source-bound.", "type": "artifact"}]

Transformation / operator

derive_formal_obligations

Bounded public warrant

At formalize_obligations, derive_formal_obligations checks ProofDeliveryPlan before emitting FormalProo fObligations. Authority boundary: May validate only the exact bound statement after the proof gate; may not silently close gaps or widen quantifiers, model, or regime.

Conclusion

formalize_obligations produces FormalProofObligations: Quarantine the proof pending the endpoint lemma.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}

Causal parents

["theorem_proof_delivery.base.formalize_obligations.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.5805d460f79924

Alternative / falsifier / failure links

alternative.3c77506adf8d81, falsifier.70680b6167757d

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["formalize_obligations-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.3 Inference 3: theorem_proof_delivery.base.build_lemma_chain.inference

---

Trace disposition

Complete public dossier

Step / skill

build_lemma_chain / lemma-chain-proof-builder

Question

At build_lemma_chain, how does FormalProofObligations bear on LemmaChain? Does the lemma DAG prove the exact quantified statement without circular support?

Typed premises

[{"statement": "FormalProofObligations: The exact theorem statement and formal obligations are source-bound.", "type": "artifact"}]

Transformation / operator

compose_lemma_dependency

Bounded public warrant

At build_lemma_chain, compose_lemma_dependency checks FormalProofObligations before emitting LemmaChain. Authority boundary: May validate only the exact bound statement after the proof gate; may not silently close gaps or widen quantifiers, model, or regime.

Conclusion

build_lemma_chain produces LemmaChain: Quarantine the proof pending the endpoint lemma.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}

Causal parents

["theorem_proof_delivery.base.build_lemma_chain.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.472f37ac7a0c04

Alternative / falsifier / failure links

alternative.8ba7ffdeb07c61, falsifier.89697243ca7906

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["build_lemma_chain-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.4 Inference 4: theorem_proof_delivery.base.inference_pvsnp_inference.inference

Trace disposition

Complete public dossier

Step / skill

control_pvsnp_inference / pvsnp-inference-quality-controller

Question

At control_pvsnp_inference, how does LemmaChain bear on InferenceQualityDecision? Does the lemma DAG prove the exact quantified statement without circular support?

Typed premises

[{"statement": "LemmaChain: The exact theorem statement and formal obligations are source-bound.", "type": "artifact"}]

Transformation / operator

artifact_grounded_transformation

Bounded public warrant

At control_pvsnp_inference, artifact_grounded_transformation checks LemmaChain before emitting InferenceQualityDecision. Authority boundary: May validate only the exact bound statement after the proof gate; may not silently close gaps or widen quantifiers, model, or regime.

Conclusion

control_pvsnp_inference produces InferenceQualityDecision: Quarantine the proof pending the endpoint lemma.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}

Causal parents

["theorem_proof_delivery.base.control_pvsnp_inference.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.3c1ec856e82082

Alternative / falsifier / failure links

alternative.812eb21648cc11, falsifier.beaf95cd1c3b15

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["control_pvsnp_inference-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

5.5 Inference 5: theorem_proof_delivery.base.gate_proof.inference

---

Trace disposition

Complete public dossier

Step / skill

gate_proof / proof-quality-gatekeeper

Question

At gate_proof, how does LemmaChain bear on ProofGateDecision? Does the lemma DAG prove the exact quantified statement without circular support?

Typed premises

[{"statement": "LemmaChain: The exact theorem statement and formal obligations are source-bound.", "type": "artifact"}]

Transformation / operator

validate_exact_claim_boundary

Bounded public warrant

At gate_proof, validate_exact_claim_boundary checks LemmaChain before emitting ProofGateDecision. Authority boundary: May validate only the exact bound statement after the proof gate; may not silently close gaps or widen quantifiers, model, or regime.

Conclusion

gate_proof produces ProofGateDecision: Quarantine the proof pending the endpoint lemma.

Uncertainty

bounded by the declared artifact and open obligations

Assumptions

None recorded

Scope before / after / model / regime

{"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}

Causal parents

["theorem_proof_delivery.base.gate_proof.control"]

Artifact integrity status

integrity_valid

Scientific epistemic status

evidence_backed

Workflow disposition

accepted

Evidence links

evidence.0e7ae18dc545b3

Alternative / falsifier / failure links

alternative.629513ec9c013d, falsifier.488b24a824f228

Obligation links

None recorded

Artifact / receipt links

None recorded

Authority

{"absent": ["unscoped-epistemic-upgrade"], "exercised": ["gate_proof-assessment"], "prohibitions_checked": ["no hidden scope promotion"]}

Handoff

None recorded

6 Alternatives, falsifiers, and failed paths

ID	Kind	Inference	Description	Disposition
alternative.fc402c24a799fd	alternative	theorem_proof_delivery.base.orchestrate_delivery.inference	Do not emit ProofDeliveryPlan: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.	rejected
falsifier.4b5d8d42fecd8b	falsifier	theorem_proof_delivery.base.orchestrate_delivery.inference	A non-circular endpoint lemma discharging the quantified obligation reopens the gate.	Not recorded
alternative.3c77506adf8d81	alternative	theorem_proof_delivery.base.formalize_obligations.inference	Do not emit FormalProofObligations: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.	rejected
falsifier.70680b6167757d	falsifier	theorem_proof_delivery.base.formalize_obligations.inference	A non-circular endpoint lemma discharging the quantified obligation reopens the gate.	Not recorded
alternative.8ba7ffdeb07c61	alternative	theorem_proof_delivery.base.build_lemma_chain.inference	Do not emit LemmaChain: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.	rejected
falsifier.89697243ca7906	falsifier	theorem_proof_delivery.base.build_lemma_chain.inference	A non-circular endpoint lemma discharging the quantified obligation reopens the gate.	Not recorded
alternative.4f700036ecee2b	alternative	theorem_proof_delivery.base.split_gap_or_countermodel.inference	Do not emit ProofGapDecision: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.	accepted
falsifier.43b378f8449feb	falsifier	theorem_proof_delivery.base.split_gap_or_countermodel.inference	A non-circular endpoint lemma discharging the quantified obligation reopens the gate.	Not recorded

ID	Kind	Inference	Description	Disposition
alternatives_and_failure.ee35355f9e96d759	failure	theorem_proof_delivery.base.split_gap_or_countermodel.inference	split_gap_or_countermodel rejects ProofGapDecision: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.	Not recorded
alternative.812eb21648cc11	alternative	theorem_proof_delivery.base.inference_pvsnp_inference.inference	Do not emit InferenceQualityDecision: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.	rejected
falsifier.beaf95cd1c3b15	falsifier	theorem_proof_delivery.base.inference_pvsnp_inference.inference	A non-circular endpoint lemma discharging the quantified obligation reopens the gate.	Not recorded
alternative.629513ec9c013d	alternative	theorem_proof_delivery.base.gate_proof.inference	Do not emit ProofGateDecision: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.	rejected
falsifier.488b24a824f228	falsifier	theorem_proof_delivery.base.gate_proof.inference	A non-circular endpoint lemma discharging the quantified obligation reopens the gate.	Not recorded

7 Cross-cutting validation ledgers

7.1 Scope transitions

Semantic ID	Complete recorded detail
theorem_proof_delivery.base.orchestrate_delivery.inference	{"event_id": "theorem_proof_delivery.base.orchestrate_delivery.inference", "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "step_id": "orchestrate_delivery"}
theorem_proof_delivery.base.formalize_obligations.inference	{"event_id": "theorem_proof_delivery.base.formalize_obligations.inference", "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "step_id": "formalize_obligations"}
theorem_proof_delivery.base.build_lemma_chain.inference	{"event_id": "theorem_proof_delivery.base.build_lemma_chain.inference", "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "step_id": "build_lemma_chain"}
theorem_proof_delivery.base.split_gap_or_countermodel.inference	{"event_id": "theorem_proof_delivery.base.split_gap_or_countermodel.inference", "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "step_id": "split_gap_or_countermodel"}
theorem_proof_delivery.base.inference_pvsnp_inference.inference	{"event_id": "theorem_proof_delivery.base.inference_pvsnp_inference.inference", "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "step_id": "control_pvsnp_inference"}
theorem_proof_delivery.base.gate_proof.inference	{"event_id": "theorem_proof_delivery.base.gate_proof.inference", "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "step_id": "gate_proof"}

7.2 Authority boundaries

Semantic ID	Complete recorded detail
theorem_proof_delivery.base.orchestrate_delivery.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "theorem_proof_delivery.base.orchestrate_delivery.inference", "exercised": ["orchestrate_delivery-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "orchestrate_delivery"}
theorem_proof_delivery.base.formalize_obligations.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "theorem_proof_delivery.base.formalize_obligations.inference", "exercised": ["formalize_obligations-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "formalize_obligations"}
theorem_proof_delivery.base.build_lemma_chain.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "theorem_proof_delivery.base.build_lemma_chain.inference", "exercised": ["build_lemma_chain-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "build_lemma_chain"}
theorem_proof_delivery.base.split_gap_or_countermodel.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "theorem_proof_delivery.base.split_gap_or_countermodel.inference", "exercised": ["split_gap_or_countermodel-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "split_gap_or_countermodel"}
theorem_proof_delivery.base.inference_pvsnp_inference.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "theorem_proof_delivery.base.inference_pvsnp_inference.inference", "exercised": ["control_pvsnp_inference-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "control_pvsnp_inference"}
theorem_proof_delivery.base.gate_proof.inference	{"absent": ["unscoped-epistemic-upgrade"], "event_id": "theorem_proof_delivery.base.gate_proof.inference", "exercised": ["gate_proof-assessment"], "prohibitions_checked": ["no hidden scope promotion"], "step_id": "gate_proof"}

7.3 Open and discharged obligations

Semantic ID	Complete recorded detail
obligation.f19f2c80a741e7	{"event_id": "theorem_proof_delivery.base.split_gap_or_countermodel.inference", "obligation_id": "obligation.f19f2c80a741e7", "requirement": "A non-circular endpoint lemma discharging the quantified obligation reopens the gate.", "status": "open"}

7.4 Handoffs

Semantic ID	Complete recorded detail
theorem_proof_delivery.handoff	{"accepted": true, "destination_workflow_id": "conceptual_hypothesis_exploration_loop", "event_id": "theorem_proof_delivery.handoff"}

A Complete chronological lifecycle ledger

This ledger records execution order. A precedes relation is not evidential support or entailment.

Event ID	Seq.	Family	Step	Disposition	Parents
theorem_proof_delivery.base.orchestrate_delivery.control	1	authorization	orchestrate_delivery	authorized	None
theorem_proof_delivery.base.orchestrate_delivery.inference	2	inference	orchestrate_delivery	accepted	["theorem_proof_delivery.base.orchestrate_delivery.control"]

Event ID	Seq.	Family	Step	Disposition	Parents
theorem_proof_delivery.base.formalize_obligations.control	3	authorization	formalize_obligations	authorized	["theorem_proof_delivery.base.orchestrate_delivery.inference"]
theorem_proof_delivery.base.formalize_obligations.inference	4	inference	formalize_obligations	accepted	["theorem_proof_delivery.base.formalize_obligations.control"]
theorem_proof_delivery.base.build_lemma_chain.control	5	authorization	build_lemma_chain	authorized	["theorem_proof_delivery.base.formalize_obligations.inference"]
theorem_proof_delivery.base.build_lemma_chain.inference	6	inference	build_lemma_chain	accepted	["theorem_proof_delivery.base.build_lemma_chain.control"]
theorem_proof_delivery.base.split_gap_or_countermodel.control	7	authorization	split_gap_or_countermodel	authorized	["theorem_proof_delivery.base.build_lemma_chain.inference"]
theorem_proof_delivery.base.split_gap_or_countermodel.inference	8	failure	split_gap_or_countermodel	failed	["theorem_proof_delivery.base.split_gap_or_countermodel.control"]
theorem_proof_delivery.base.control_pvsnp_inference.control	9	authorization	control_pvsnp_inference	authorized	["theorem_proof_delivery.base.split_gap_or_countermodel.inference"]
theorem_proof_delivery.base.inference_pvsnp_inference.inference	10	inference	control_pvsnp_inference	accepted	["theorem_proof_delivery.base.control_pvsnp_inference.control"]
theorem_proof_delivery.base.gate_proof.control	11	authorization	gate_proof	authorized	["theorem_proof_delivery.base.inference_pvsnp_inference.inference"]
theorem_proof_delivery.base.gate_proof.inference	12	inference	gate_proof	accepted	["theorem_proof_delivery.base.gate_proof.control"]
theorem_proof_delivery.handoff	13	handoff	Not recorded	handed_off	["theorem_proof_delivery.base.gate_proof.inference"]

B Claim ledger

Semantic ID	Complete recorded detail
theorem_proof_delivery.base.orchestrate_delivery.inference	{"claim_id": "theorem_proof_delivery.base.orchestrate_delivery.inference", "epistemic_status": "evidence_backed", "event_id": "theorem_proof_delivery.base.orchestrate_delivery.inference", "evidence_ids": ["evidence.5dba134bef7cd8"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "text": "orchestrate_delivery produces ProofDeliveryPlan: Quarantine the proof pending the endpoint lemma.", "trace_incomplete": false, "workflow_disposition": "accepted"}

Semantic ID	Complete recorded detail
theorem_proof_delivery.base.formalize_obligations.inference	{"claim_id": "theorem_proof_delivery.base.formalize_obligations.inference", "epistemic_status": "evidence_backed", "event_id": "theorem_proof_delivery.base.formalize_obligations.inference", "evidence_ids": ["evidence.5805d460f79924"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "text": "formalize_obligations produces FormalProofObligations: Quarantine the proof pending the endpoint lemma.", "trace_incomplete": false, "workflow_disposition": "accepted"}
theorem_proof_delivery.base.build_lemma_chain.inference	{"claim_id": "theorem_proof_delivery.base.build_lemma_chain.inference", "epistemic_status": "evidence_backed", "event_id": "theorem_proof_delivery.base.build_lemma_chain.inference", "evidence_ids": ["evidence.472f37ac7a0c04"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "text": "build_lemma_chain produces LemmaChain: Quarantine the proof pending the endpoint lemma.", "trace_incomplete": false, "workflow_disposition": "accepted"}
theorem_proof_delivery.base.split_gap_or_countermodel.inference	{"claim_id": "theorem_proof_delivery.base.split_gap_or_countermodel.inference", "epistemic_status": "quarantined", "event_id": "theorem_proof_delivery.base.split_gap_or_countermodel.inference", "evidence_ids": ["evidence.a50c5cd202fe28"], "integrity_status": "integrity_valid", "obligation_ids": ["obligation.f19f2c80a741e7"], "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "text": "split_gap_or_countermodel rejects ProofGapDecision: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.", "trace_incomplete": false, "workflow_disposition": "failed"}
theorem_proof_delivery.base.inference_pvsnp_inference.inference	{"claim_id": "theorem_proof_delivery.base.inference_pvsnp_inference.inference", "epistemic_status": "evidence_backed", "event_id": "theorem_proof_delivery.base.inference_pvsnp_inference.inference", "evidence_ids": ["evidence.3c1ec856e82082"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "text": "control_pvsnp_inference produces InferenceQualityDecision: Quarantine the proof pending the endpoint lemma.", "trace_incomplete": false, "workflow_disposition": "accepted"}
theorem_proof_delivery.base.gate_proof.inference	{"claim_id": "theorem_proof_delivery.base.gate_proof.inference", "epistemic_status": "evidence_backed", "event_id": "theorem_proof_delivery.base.gate_proof.inference", "evidence_ids": ["evidence.0e7ae18dc545b3"], "integrity_status": "integrity_valid", "obligation_ids": [], "scope": {"after": "quarantined lemma DAG", "before": "proof queue item", "model": "declared source model", "regime": "exact theorem"}, "text": "gate_proof produces ProofGateDecision: Quarantine the proof pending the endpoint lemma.", "trace_incomplete": false, "workflow_disposition": "accepted"}

C Typed evidence ledger

Semantic ID	Complete recorded detail
evidence.5dba134bef7cd8	{"artifact_type": "ProofQueueItem", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "theorem_proof_delivery.base.orchestrate_delivery.inference", "evidence_id": "evidence.5dba134bef7cd8", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.5805d460f79924	{"artifact_type": "ProofDeliveryPlan", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "theorem_proof_delivery.base.formalize_obligations.inference", "evidence_id": "evidence.5805d460f79924", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}

Semantic ID	Complete recorded detail
evidence.472f37ac7a0c04	{"artifact_type": "FormalProofObligations", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "theorem_proof_delivery.base.build_lemma_chain.inference", "evidence_id": "evidence.472f37ac7a0c04", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.a50c5cd202fe28	{"artifact_type": "LemmaChain", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "theorem_proof_delivery.base.split_gap_or_countermodel.inference", "evidence_id": "evidence.a50c5cd202fe28", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "refutes"}
evidence.3c1ec856e82082	{"artifact_type": "LemmaChain", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "theorem_proof_delivery.base.inference_pvsnp_inference.inference", "evidence_id": "evidence.3c1ec856e82082", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}
evidence.0e7ae18dc545b3	{"artifact_type": "LemmaChain", "digest": "61c8ddbab2b6d4dded3cbe6cf9348c5fd6c658796985c80675774399043f4a90", "event_id": "theorem_proof_delivery.base.gate_proof.inference", "evidence_id": "evidence.0e7ae18dc545b3", "path": "tests/fixtures/reasoning_trace/evidence/workflow-evidence.json", "relation": "supports"}

D Decision and gate ledger

Semantic ID	Complete recorded detail
theorem_proof_delivery.base.orchestrate_delivery.control	{"disposition": "authorized", "event_id": "theorem_proof_delivery.base.orchestrate_delivery.control", "family": "authorization", "reason": "", "step_id": "orchestrate_delivery"}
theorem_proof_delivery.base.formalize_obligations.control	{"disposition": "authorized", "event_id": "theorem_proof_delivery.base.formalize_obligations.control", "family": "authorization", "reason": "", "step_id": "formalize_obligations"}
theorem_proof_delivery.base.build_lemma_chain.control	{"disposition": "authorized", "event_id": "theorem_proof_delivery.base.build_lemma_chain.control", "family": "authorization", "reason": "", "step_id": "build_lemma_chain"}
theorem_proof_delivery.base.split_gap_or_countermodel.control	{"disposition": "authorized", "event_id": "theorem_proof_delivery.base.split_gap_or_countermodel.control", "family": "authorization", "reason": "", "step_id": "split_gap_or_countermodel"}
theorem_proof_delivery.base.control_pvsnp_inference.control	{"disposition": "authorized", "event_id": "theorem_proof_delivery.base.control_pvsnp_inference.control", "family": "authorization", "reason": "", "step_id": "control_pvsnp_inference"}
theorem_proof_delivery.base.gate_proof.control	{"disposition": "authorized", "event_id": "theorem_proof_delivery.base.gate_proof.control", "family": "authorization", "reason": "", "step_id": "gate_proof"}

E Artifact and receipt ledger

Semantic ID	Complete recorded detail
No artifact and receipt ledger recorded.	

F Specialized projection ledger

Semantic ID	Complete recorded detail
No specialized projection ledger recorded.	

G Causal and workflow graph ledger

Typed edge summary

precedes: 17, contains: 13, supports: 5, refutes: 1, produces: 1

Edge meanings are explicit: precedes is chronology; supports/refutes/constrains are epistemic relations; contains/produces/branches/joins/refines/validates/blocks/hands-off are workflow relations.

G.1 Nodes

Node	Type	Label	Status
theorem_proof_delivery.base.orchestrate_delivery.control	event	authorization	Not recorded
theorem_proof_delivery.base.orchestrate_delivery.inference	event	orchestrate_delivery produces ProofDeliveryPlan: Quarantine the proof pending the endpoint lemma.	Not recorded
evidence.5dba134bef7cd8	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
theorem_proof_delivery.base.formalize_obligations.control	event	authorization	Not recorded
theorem_proof_delivery.base.formalize_obligations.inference	event	formalize_obligations produces FormalProofObligations: Quarantine the proof pending the endpoint lemma.	Not recorded
evidence.5805d460f79924	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
theorem_proof_delivery.base.build_lemma_chain.control	event	authorization	Not recorded
theorem_proof_delivery.base.build_lemma_chain.inference	event	build_lemma_chain produces LemmaChain: Quarantine the proof pending the endpoint lemma.	Not recorded
evidence.472f37ac7a0c04	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
theorem_proof_delivery.base.split_gap_or_countermodel.control	event	authorization	Not recorded
theorem_proof_delivery.base.split_gap_or_countermodel.inference	event	split_gap_or_countermodel rejects ProofGapDecision: A countermodel invalidates the proposed endpoint lemma and prevents silent gap closure.	Not recorded
evidence.a50c5cd202fe28	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
obligation.f19f2c80a741e7	obligation	A non-circular endpoint lemma discharging the quantified obligation reopens the gate.	Not recorded
theorem_proof_delivery.base.control_pvsnp_inference.control	event	authorization	Not recorded
theorem_proof_delivery.base.inference_pvsnp_inference.inference	event	control_pvsnp_inference produces InferenceQualityDecision: Quarantine the proof pending the endpoint lemma.	Not recorded
evidence.3c1ec856e82082	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded

Node	Type	Label	Status
theorem_proof_delivery.base.gate_proof.control	event	authorization	Not recorded
theorem_proof_delivery.base.gate_proof.inference	event	gate_proof produces ProofGateDecision: Quarantine the proof pending the endpoint lemma.	Not recorded
evidence.0e7ae18dc545b3	evidence	tests/fixtures/reasoning_trace/evidence/workflow-evidence.json	Not recorded
theorem_proof_delivery.handoff	event	Typed handoff to conceptual_hypothesis_exploration_loop	Not recorded
step:orchestrate_delivery	step	orchestrate_delivery	observed
step:formalize_obligations	step	formalize_obligations	observed
step:build_lemma_chain	step	build_lemma_chain	observed
step:split_gap_or_countermodel	step	split_gap_or_countermodel	observed
step:control_pvsnp_inference	step	control_pvsnp_inference	observed
step:gate_proof	step	gate_proof	observed

G.2 Complete typed edge table

Source	Relation	Target	Adapter / detail
theorem_proof_delivery.base.orchestrate_delivery.control	precedes	theorem_proof_delivery.base.orchestrate_delivery.inference	None
evidence.5dba134bef7cd8	supports	theorem_proof_delivery.base.orchestrate_delivery.inference	None
theorem_proof_delivery.base.orchestrate_delivery.inference	precedes	theorem_proof_delivery.base.formalize_obligations.control	None
theorem_proof_delivery.base.formalize_obligations.control	precedes	theorem_proof_delivery.base.formalize_obligations.inference	None
evidence.5805d460f79924	supports	theorem_proof_delivery.base.formalize_obligations.inference	None
theorem_proof_delivery.base.formalize_obligations.inference	precedes	theorem_proof_delivery.base.build_lemma_chain.control	None
theorem_proof_delivery.base.build_lemma_chain.control	precedes	theorem_proof_delivery.base.build_lemma_chain.inference	None
evidence.472f37ac7a0c04	supports	theorem_proof_delivery.base.build_lemma_chain.inference	None
theorem_proof_delivery.base.build_lemma_chain.inference	precedes	theorem_proof_delivery.base.split_gap_or_countermodel.control	None
theorem_proof_delivery.base.split_gap_or_countermodel.control	precedes	theorem_proof_delivery.base.split_gap_or_countermodel.inference	None
evidence.a50c5cd202fe28	refutes	theorem_proof_delivery.base.split_gap_or_countermodel.inference	None
theorem_proof_delivery.base.split_gap_or_countermodel.inference	produces	obligation.f19f2c80a741e7	None
theorem_proof_delivery.base.split_gap_or_countermodel.inference	precedes	theorem_proof_delivery.base.control_pvsnp_inference.control	None

Source	Relation	Target	Adapter / detail
theorem_proof_delivery.base.control_pvsnp_inference.control	precedes	theorem_proof_delivery.base.inference_pvsnp_inference.inference	None
evidence.3c1ec856e82082	supports	theorem_proof_delivery.base.inference_pvsnp_inference.inference	None
theorem_proof_delivery.base.inference_pvsnp_inference.inference	precedes	theorem_proof_delivery.base.gate_proof.control	None
theorem_proof_delivery.base.gate_proof.control	precedes	theorem_proof_delivery.base.gate_proof.inference	None
evidence.0e7ae18dc545b3	supports	theorem_proof_delivery.base.gate_proof.inference	None
theorem_proof_delivery.base.gate_proof.inference	precedes	theorem_proof_delivery.handoff	None
step:orchestrate_delivery	contains	theorem_proof_delivery.base.orchestrate_delivery.control	None
step:orchestrate_delivery	contains	theorem_proof_delivery.base.orchestrate_delivery.inference	None
step:formalize_obligations	contains	theorem_proof_delivery.base.formalize_obligations.control	None
step:formalize_obligations	contains	theorem_proof_delivery.base.formalize_obligations.inference	None
step:build_lemma_chain	contains	theorem_proof_delivery.base.build_lemma_chain.control	None
step:build_lemma_chain	contains	theorem_proof_delivery.base.build_lemma_chain.inference	None
step:split_gap_or_countermodel	contains	theorem_proof_delivery.base.split_gap_or_countermodel.control	None
step:split_gap_or_countermodel	contains	theorem_proof_delivery.base.split_gap_or_countermodel.inference	None
step:split_gap_or_countermodel	contains	theorem_proof_delivery.base.split_gap_or_countermodel.inference	None
step:control_pvsnp_inference	contains	theorem_proof_delivery.base.control_pvsnp_inference.control	None
step:control_pvsnp_inference	contains	theorem_proof_delivery.base.inference_pvsnp_inference.inference	None
step:gate_proof	contains	theorem_proof_delivery.base.gate_proof.control	None
step:gate_proof	contains	theorem_proof_delivery.base.gate_proof.inference	None
step:orchestrate_delivery	precedes	step:formalize_obligations	None
step:formalize_obligations	precedes	step:build_lemma_chain	None
step:build_lemma_chain	precedes	step:split_gap_or_countermodel	None
step:split_gap_or_countermodel	precedes	step:control_pvsnp_inference	None
step:control_pvsnp_inference	precedes	step:gate_proof	None

H Projection completeness and omission ledger

Packet field	Disposition	Destination	Reason
alternatives_and_failures	rendered	negative-path body	Not recorded
artifacts_and_receipts	summarized_with_complete_appendix	artifact appendix	Not recorded
authority_ledger	rendered	authority body and appendix	Not recorded
claims	summarized_with_complete_appendix	claim ledger appendix	Not recorded
coverage	rendered	validation dashboard	Not recorded
decisions_and_gates	summarized_with_complete_appendix	decision and gate appendix	Not recorded
declared_route	rendered	workflow route body	Not recorded
evidence	summarized_with_complete_appendix	evidence ledger appendix	Not recorded
graph	summarized_with_complete_appendix	graph summary and complete edge appendix	Not recorded
handoffs	rendered	handoff body	Not recorded
identity	rendered	cover identity	Not recorded
inferences	rendered	expert inference dossiers	Not recorded
lens	rendered	cover and print profile	Not recorded
obligations	rendered	obligation body and appendix	Not recorded
profile_topology	rendered	route topology body	Not recorded
schema_version	rendered	build provenance	Not recorded
scope_ledger	rendered	scope body and appendix	Not recorded
specialized_projections	summarized_with_complete_appendix	specialized projection appendix	Not recorded
timeline	summarized_with_complete_appendix	complete chronology appendix	Not recorded

I Build provenance

Source packet lens

{"accent": "#64d68a", "focus_question": "Do the obligations and lemma chain establish the exact theorem?", "labels": {"claims": "Theorem and lemmas", "failures": "Gaps and countermodels", "gates": "Proof quality gate", "obligations": "Proof obligations"}, "lens_id": "theorem_proof_delivery", "mandatory_spine": ["orchestrate_delivery", "formalize_obligations", "build_lemma_chain", "gate_proof"], "metrics": ["semantic_coverage", "evidence_resolution", "open_obligations", "trace_gaps"], "required_views": ["route", "inferences", "claims", "failures", "scope", "authority", "handoffs", "coverage"], "spine_label": "Statement to proof gate", "visible_name": "Theorem Proof Delivery"}

Packet SHA-256

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Renderer version

1.0

Print inventory schema

1.0